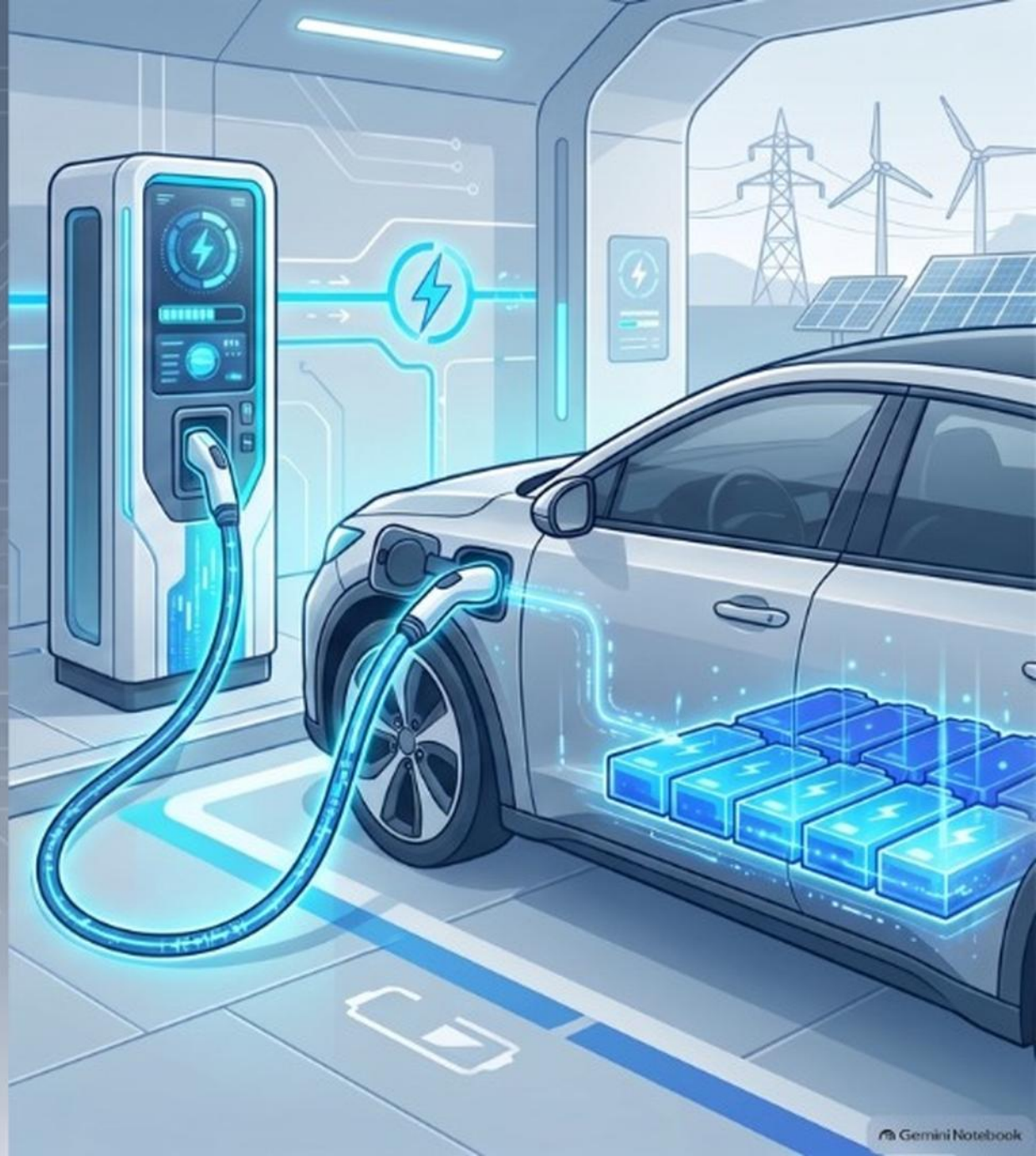
The Physics of Ultra-Rapid EV Charging

GCSE Physics (9-1) · Electricity, Energy Transfer & High-Power Engineering

- **Electrical Power** ($P = IV$ and $P = I^2R$)
- **Charge Flow & Energy** ($Q = It$ and $E = QV$)
- **Energy Transfer Rate** ($P = E / t$)



Gemini Notebook

Activating Your Physics Toolkit

Retrieve the baseline equations and metrics required for high-power electrical engineering.

What formula links electrical charge (Q), current (I), and time (t)?

[Success Cue]: Q = I × t

Q = I × t (Coulombs = Amps × seconds)

What is the unit of electrical power, and what does 1 Watt equal in energy per second?

[Success Cue]: Joules per second

Watt (W) = 1 Joule per second (1 J/s)

What equation links energy transferred (E), charge (Q), and potential difference (V)?

[Success Cue]: E = Q × V

E = Q × V (Joules = Coulombs × Volts)

A domestic kettle operates on 230 V mains and draws 10 A. Calculate its power rating.

[Success Cue]: P = V × I

2,300 W (2.3 kW)

How much electrical energy is transferred when a 2.3 kW kettle boils water for 200 seconds?

[Success Cue]: E = P × t

460,000 J (460 kJ)

If electrical current through a charging cable doubles, by what factor does resistive heat loss increase?

[Success Cue]: P = I^2 × R squared relationship

Power quadruples (increases 4×)

Gemini Notebook

The 152 Kettle Puzzle

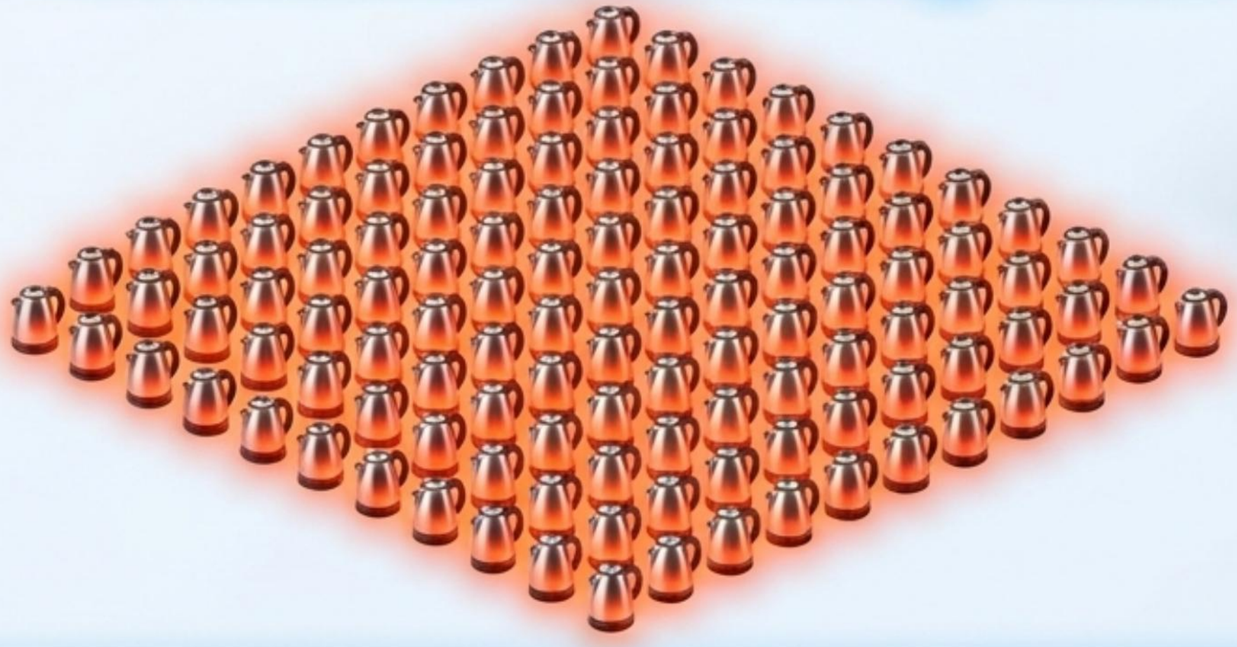
A standard domestic kitchen kettle draws 2,300 W. An ultra-rapid EV supercharger delivers 350,000 W. How many boiling kettles of electrical power are flowing simultaneously through that single charging cable?

=

~152 Simultaneous Boiling Kettles



2,300 Watts
(2.3 kW)





350,000 Watts
(350 kW)

How do engineers deliver the power of 152 kettles through a single handheld cable without melting the cable or destroying the battery?

Gemini Notebook

ONE MONTH OF PREMIUM AI USE



0.69 kWh

ILLUSTRATIVE SCENARIO: 600 REQUESTS / MONTH

≈ 6.3
kettle boils
1 litre each

≈ 46
smartphone charges
15 Wh each

≈ 1.2%
of an EV battery
60 kWh battery

Illustrative scenario, not a measured subscriber average

Power Dictates Charging Speed

Power determines how fast energy is transferred to the battery, not the total battery capacity.

P = E / t (Power in Watts = Energy transferred in Joules ÷ Time taken in seconds)



Domestic 3-Pin Plug

2.3 kW

~32.6 hours
(Overnight + Full Day)

5%



Home Wallbox

7.4 kW

~10.1 hours
(Single Overnight Charge)

50%



Motorway Rapid DC Charger

150 kW

~30 minutes
(Quick Coffee Stop)

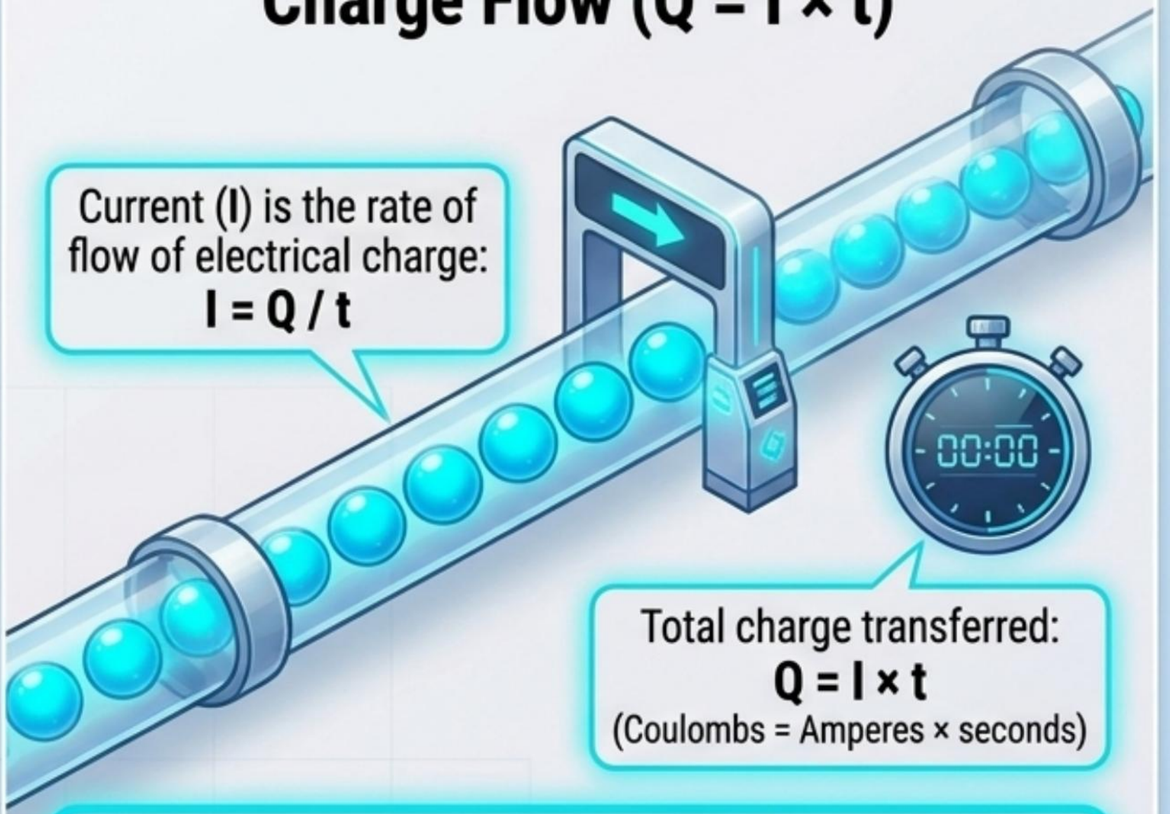
95%

Gemini Notebook

The Physics Inside the Battery

How electrical charge carriers deliver massive energy payloads into the battery cells.

Charge Flow (Q = I × t)



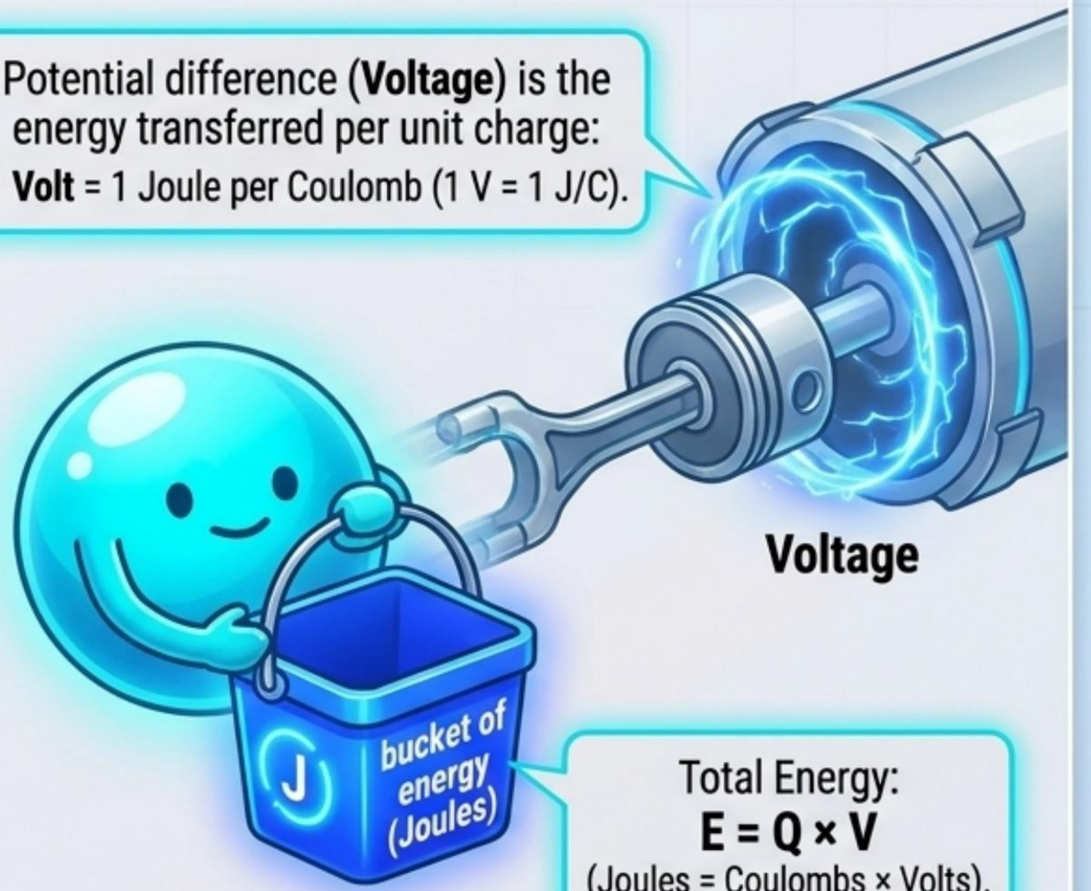
Current (**I**) is the rate of flow of electrical charge:
I = Q / t

Total charge transferred:
Q = I × t
(Coulombs = Amperes × seconds)

EV Real-World Metric: An EV charging at 200 A for 30 minutes (1,800 s) receives **Q = 200 A × 1,800 s = 360,000 Coulombs (C)** of electrical charge!

Energy per Charge (E = Q × V)

Potential difference (**Voltage**) is the energy transferred per unit charge:
1 Volt = 1 Joule per Coulomb (1 V = 1 J/C).



Total Energy:
E = Q × V
(Joules = Coulombs × Volts).

Gemini Notebook

The Heat Trap: The Physics of Dissipated Energy

Cables have resistance (R). Forcing current (I) through resistance causes energy to be wasted as heat. We cannot simply pump infinite current to charge cars faster.



$$P = I^2R$$

(Power lost as heat = Current squared × Resistance)

The Squared Effect: Because current is squared, doubling the current doesn't double the heat—it **quadruples** it. Pumping 500 Amps through a standard cable would melt the rubber insulation and cause a fire.

Gemini Notebook

The Voltage Solution: Outsmarting Resistance

Delivering massive power without massive heat. If P = I × V, we can reach our 350,000 W Supercharger target in two entirely different ways.

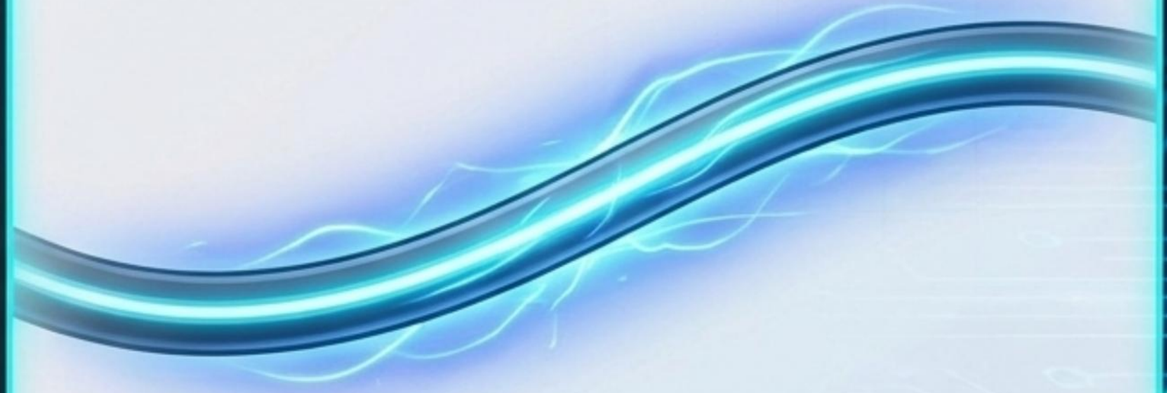
400V System
(The Old Standard)



875 Amps × 400 Volts = **350,000 W**

⚠ **Result:** Cable is too heavy to lift, and I^2R heat dissipation melts the casing.

800V System
(The Modern Tech, e.g., Porsche Taycan)



437 Amps × **800 Volts** = **350,000 W**

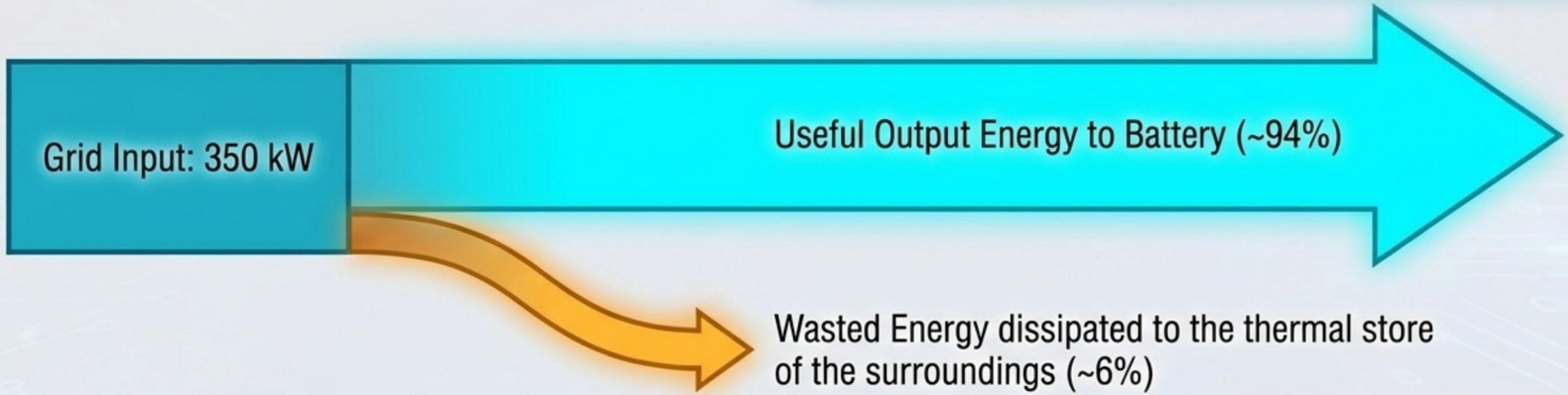
✓ **Result:** Same power delivered. Half the current. 4x less heat generated.

Gemini Notebook

Where Does the Lost Energy Go?

The law of conservation of energy. A supercharger draws 350kW from the national grid, but the car battery might only receive 330kW.

Efficiency = (Useful Output Energy ÷ Total Input Energy)



Real-World Application: Despite moving to 800V, some I^2R heat is unavoidable at 350kW. Engineers use liquid-cooled charging cables, pumping coolant directly through the cable jacket to absorb dissipated heat, maintaining safety and maximizing transfer efficiency.

Gemini Notebook

The Examiner's Trap: Units & Significant Figures

How to secure maximum marks on high-value 6-mark calculation questions by avoiding common OCR examiner pitfalls.

Time Conversions	~~Minutes / Hours~~	**Seconds (s)**
	Example: 45 mins must become 45 × 60 = 2,700 s	
Power Conversions	~~Kilowatts (kW)~~	**Watts (W)**
	Example: 150 kW must become 150,000 W for standard equations	
Energy Units	Standard scientific unit is Joules (J). Do not confuse calculating mechanical/electrical Joules with reading a domestic utility bill in kWh.	
Significant Figures	Warning: Always round your final answer to the lowest number of significant figures given in the question data. (e.g., if current is given as 120 A (2 sig figs), the final answer must be 2 sig figs).	

Gemini Notebook

The EV Charging Network Matrix

Synthesizing real-world physics metrics across the charging ecosystem.

	Voltage (V)	Current (I)	Power (P = IV)	Typical Charge Time (t)
Home Wallbox	230 V	32 A	7.4 kW	10 hours
Standard Rapid Charger	400 V	125 A	50 kW	1 hour
Ultra-Rapid Architecture (The Pinnacle)	800 V	437 A	350 kW	15 mins

Gemini Notebook

Step-by-Step: The 6-Mark Exam Challenge

An EV charger operates at 400 V and delivers 120 A of current for 45 minutes. Calculate the total energy transferred to the battery in Joules. Give your answer to 2 significant figures.

Step 1: The Trap (Conversions)

⚠ t = 45 mins × 60 = 2,700 s

Step 2: Power Equation

$P = I \times V$

400 V × 120 A = 48,000 W

Step 3: Energy Equation

$E = P \times t$

48,000 W × 2,700 s = 129,600,000 J

Step 4: Final Polish (Sig Figs)

Convert 129,600,000 J to 2 significant figures.

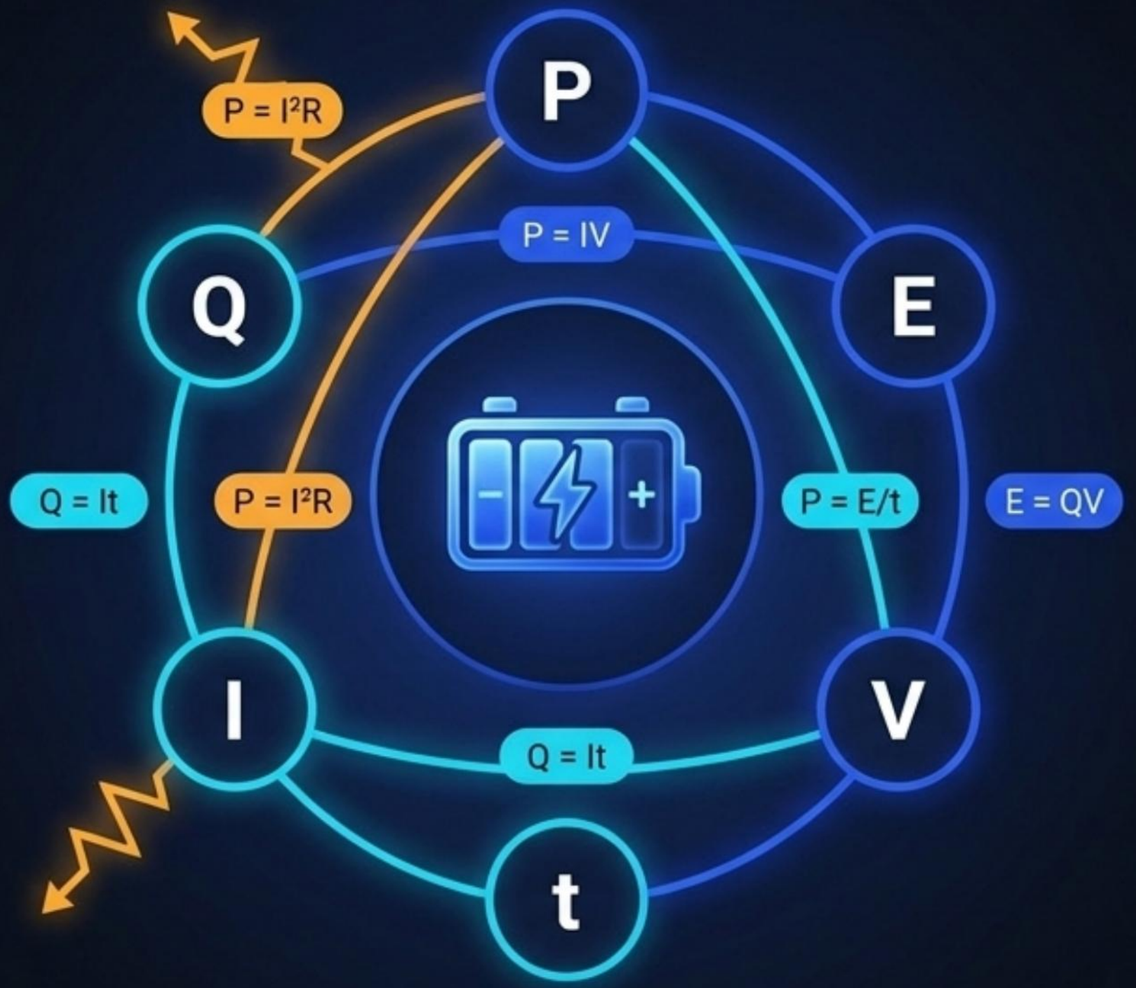
Answer: 1.3 × 10^8 J (or 130,000,000 J)

Alternatively, calculate $Q = It$ (324,000 C), then $E = QV$ to arrive at the exact same answer, proving the formulas interlock.

Gemini Notebook

The Engineering Toolkit

Physics isn't about memorizing isolated rules; it's about understanding how energy, charge, and resistance interlock to shape the modern world.



Gemini Notebook